Mark Cheli

Director, Product Operations & Strategy · Onshape (PTC)

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I steer how Onshape's R&D organization plans, measures and builds: the operating cadence, the adoption of AI across product development, the analytics leadership plans against, and the strategy work that sets direction. I am a builder by nature who now spends most days steering the organization — and I still prototype the answer before asking for it. Thirteen years at PTC from intern to director, including engineering teams of 35+ and a 40-person training-products organization. At home: a 40-container homelab and the apps that run on it.

EXPERIENCE

Director, Product Operations & Strategy — Jan 2022 – present

PTC — Onshape · Boston, MA

Run the team that decides how Onshape's R&D organization plans, measures and builds. I own the operating cadence (annual epic planning, reviews, tech-debt planning), lead the move to AI-assisted software development, and own the product analytics that leadership and go-to-market plan against. Hands-on where it is faster: I write the first version of internal tools myself.

- Leading Onshape's move to AI-assisted development: the agentic-harness guidelines engineers follow, the evaluation of Claude over AWS Bedrock for 1,000+ users, MCP servers that put the data warehouse and data dictionary in front of every agent, and the governance and budget model for AI tooling across PTC.
- Own the R&D operating cadence — annual epic planning, planning reviews, tech-debt planning — and the strategy work around it: the product-led growth discovery program, the financial case for Onshape Government, and the API quota and rate-limit policy tied to plan tiers.
- Built and run the product analytics platform the organization plans against (AWS Redshift, dbt, Prefect, Streamlit, Looker), from signup funnels to API usage.
- Run localization operations for an 11-locale product: translation-management vendor selection and renegotiation, the XLIFF/TBX pipeline, a 1,300-term glossary.
- Still ship production code: internal Flask services connecting Onshape to Salesforce, billing and the warehouse; a Python admin library; the internal product and admin portals.
- Interim Director of Business Operations for the Onshape business unit in 2022 (lead routing, CPQ, executive dashboards). Hire and manage a product-ops/analytics team, developers and interns.

Director, Education Product Feb 2021 – Jan 2022

PTC · Boston, MA

Led product for PTC's Education organization: services, support, partnerships, community and marketing for the academic programs that put Onshape, Creo and ThingWorx into schools and universities.

Senior Manager, Education Experience Jan 2020 – Feb 2021

PTC · Boston, MA

Owned the end-to-end experience for PTC's education customers after the Onshape acquisition.

Senior Manager, Customer Training Technology & Research

Jan 2019 – Jan 2020

PTC · Boston, MA

Head of product for PTC's customer training: eLearning, virtual instructor-led training and in-product guidance. Led a 40+ person organization across the US and Pune, India.

Senior Software Development Manager, PTC University EdTech R&D

May 2018 – Jan 2019

PTC · Needham, MA

Managed 35+ engineers in three teams building PTC's learning platforms.

Technology Lead, Academic & Customer Training Products

Apr 2017 – May 2018

PTC · Needham, MA

Lead engineer and product manager for IoTU.com and the academic product line, with teams in the US and India.

Senior Full-Stack Software Engineer Oct 2016 – Apr 2017

PTC · Needham, MA

Led the Academic Software Development group. IoT education products on the MEAN stack and ThingWorx.

Academic Software Engineer May 2013 – Oct 2016

PTC · Needham, MA

Joined as a student intern while finishing my degree, then full-time. Built curriculum, demos and applications for Creo, Windchill, Mathcad and ThingWorx.

Web Developer Dec 2012 – Dec 2014

Freelance

Small-business websites: HTML/CSS, PHP/MySQL, WordPress.

Assistant Program Facilitator Dec 2011 – May 2013

Christa McAuliffe Center / Challenger Learning Center, Framingham State University

Ran planetarium shows and simulated space missions for middle-school groups; handled the center's IT.

EDUCATION

B.S., Business & Information Technology 2011 – 2015

Framingham State University · Summa Cum Laude, GPA 3.94

CERTIFICATES

- Certified Onshape Associate · Onshape · 2021-01

SKILLS

Steering an R&D organization	Operating cadence & annual planning · Product strategy · AI adoption across the SDLC · Localization operations · Hiring & management
Data	SQL · AWS Redshift · dbt · Prefect · Streamlit · Looker
Building	Python (Flask, FastAPI) · TypeScript (Vue, Nuxt) · PostgreSQL · Docker · GitHub Actions · Linux
AI tooling	Agentic coding harnesses · MCP servers · Claude Code · GitHub Copilot · LiteLLM on AWS Bedrock

INTERESTS

Homelab & self-hosting	Dell PowerEdge R630 · Docker · Observability
Sim racing & flight sim	iRacing · DCS · RigReady
Old trucks	1991 Honda Acty · 1997 GMC Suburban · 1996 Mazda B4000
Outdoors	Backpacking · Car camping · Ham radio
Making	3D printing · Welding · Woodworking

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